Research in Mathematics: Discrete Mathematics

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Declaration

I hereby declare that this project report is my own work and has not been submitted to any other university or institution for any degree or diploma. All sources used are acknowledged in the bibliography.

Acknowledgement

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Chapter 1

Logic and Sets

1.1 Introduction

This chapter introduces propositional logic. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

1.2 Main Content

We study logical connectives, truth tables and sets. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

1.3 Examples

The statement $p \to q$ is false only when p is true and q is false. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

Chapter 2

Relations and Functions

2.1 Introduction

This chapter introduces relations on a set. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

2.2 Main Content

We study equivalence relations and functions. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

2.3 Examples

Congruence modulo n is an equivalence relation on the integers. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

Chapter 3

Counting and Combinatorics

3.1 Introduction

This chapter introduces the basic principles of counting. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

3.2 Main Content

We study permutations and combinations. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

3.3 Examples

The number of ways to choose r objects from n is $\binom{n}{r}$. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

Chapter 4

Graph Theory

4.1 Introduction

This chapter introduces graphs and their basic properties. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

4.2 Main Content

We study degrees, paths and trees. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

4.3 Examples

A tree with n vertices has exactly $n - 1$ edges. This topic is discussed in Rosen [1] and forms a basic part of discrete mathematics.

Chapter 5

Conclusion

In this report we studied the basic ideas of discrete mathematics, namely logic, relations, counting and graphs, following Rosen [1]. These ideas form the foundation for further study in computer science and combinatorics.

Bibliography

[1] K. H. Rosen, *Discrete Mathematics and Its Applications*, McGraw-Hill.